Finite Correlation Length and Propagation Constraints in Adaptive Substrates

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March 9, 2026

Abstract

Missing axis: Papers 1 and 2 establish geometric and kinetic laws for capacity and bandwidth. This paper completes the trilogy by isolating the propagation constraint: how far can information travel through a viability-gated system?
Central finding: Locally-coupled information systems have a *finite correlation length*. The copy-forward relay (wave → collapse → birth → fieldM seeding) is inherently local, and zone structure degrades with scale: normalized coherence length decreases 5× over a 4× scaling of system size (V108). This hard propagation ceiling — not a property specific to the lattice implementation, but a consequence of local coupling — explains why spatial structure emerges robustly at moderate scales (V100–V102) but fails at large scales (V94).
Scaling law: Coherence length scales with coupling-to-noise ratio as $L \approx C \cdot \sqrt{\kappa/\nu}$ ($C \approx 0.19$ measured; group-mean R^2=0.81, V104 diffusion sweep). Group-mean R^2 is high (0.81, r=0.90) because the law governs expected-value behavior; individual-seed R^2 is lower (0.24) due to bimodal relay outcomes (each seed either succeeds or fails stochastically). V105 (consolidation speed) shows no independent effect and is reported separately as a null result.
Evidence: Five experiments isolate field diffusion (V104), consolidation speed (V105), turnover noise amplitude (V106), relay depth (V107), and scale invariance (V108). Relay sustains to depth 5 at all coupling levels, with differentiated profiles (weak 1.13×, std 1.14×, strong 0.93×) now visible after correcting a constructor parameter bug.
Implication: The finite correlation length is the third independent constraint on viability-gated systems, complementing capacity (Paper 1) and bandwidth (Paper 2). Regional coupling that bypasses the correlation length limit (geographic relay, V100–V102) is the architectural response — not a hack, but the correct engineering consequence of the constraint.

1 Introduction

Previous work established geometric and kinetic constraints on viability-gated information systems [2, 3]. This paper focuses on the complementary constraint: **propagation** — what limits how far and how reliably information can travel through such systems?

The research program is built on three independent experimental observations, each revealing an invariant scaling law:

*Code and papers: `https://github.com/AccidentalGenius101/adaptive-memory-theory`

Axis	Question	Law / Constraint	Parameter
Paper 1: GEOMETRY	How many states?	$N \propto \tau^{-d_{\text{seen}}}$	Separation τ
Paper 2: KINETICS	How fast?	$B \propto K/\tau_{\text{hl}}$	Half-life τ_{hl}
Paper 3: PROPAGATION	How far?	$L \lesssim C\sqrt{\kappa/\nu}$ (group-mean R^2=0.81)	Coupling κ, noise ν

These three axes correspond to independent constraints on viability-gated information substrates: representational capacity (geometry), temporal adaptation rate (kinetics), and spatial transmission limits (propagation).

1.1 Mechanism and substrate variants

The three laws (geometry, kinetics, propagation) describe a general mechanism, not a single architecture. VCSM instantiates this mechanism in one specific substrate (the coupled-map lattice, called VCSM-L). This paper and related work demonstrate that the same geometric and kinetic laws, and the propagation constraints described here, hold across multiple distinct substrates:

1. **VCSM-L (Lattice):** The primary substrate in this trilogy. Information propagates through local diffusion and intergenerational memory on a 2D grid.

2. **VCSM-G (Geographic relay):** Multiple independent VCSM-L regions coupled through geographic proximity and shared wave targets (the relay experiments in Section 2).

3. **VCSM-E (Real embeddings):** The routing rule applied directly to embeddings from sentence-transformers (SBERT), bypassing lattice structure entirely.

The existence of multiple substrates implementing the same three laws suggests the principles are consequences of the geometric routing mechanism itself (VGRT: Viability-Gated Routing with Turnover), not of lattice-specific properties. This paper completes the theoretical frame by isolating the propagation constraint, the third law governing information flow through viability-gated systems.

This paper answers the third question: **How far and how reliably can information travel through a system with mandatory turnover?**

The question is non-trivial because information cannot propagate through global optimization (there is none—the system is locally viability-gated). Yet the geographic relay experiments show that structure *does* propagate across multiple coupled regions, provided they are organized geographically.

The key insight: information propagates through *spatial organization*, not through learning rules or gradient descent.

1.2 What propagation means

In standard neural networks, "propagation" typically means gradient flow or information flow through layers. Here, we mean something different: **how far a representational structure established in one region can coherently influence the structure in distant regions, given only local coupling and viability constraints.**

Specifically, we ask:

1. Does zone-structured activity in Region A propagate to Regions B, C, D?
2. How many regions deep can the relay persist before signal degrades?
3. What coupling strength is required to sustain relay at depth D?
4. Does the propagation constraint scale invariantly across system sizes?

1.3 Contrast to gradient-based propagation

In backpropagation, information propagates backward through the Jacobian of each layer, with exponential amplification (or decay) determined by the singular values of the weight matrices. Gradients can vanish or explode over depth.

In VCML relay, information propagates forward through *spatial diffusion and reinitialization*. The mechanism is:

1. Region A develops zone-structured centroids (from waves targeting specific zones)
2. Region A's fieldM values diffuse to neighbors at low rate (`FIELD_DIFFUSE_RATE` ≈ 0.02)
3. New slots in Region B are seeded from the diffused fieldM of neighbors (intergenerational memory)
4. Thus new slots inherit the zone structure, creating structured activity in Region B
5. This repeats for Region C, D, etc.

The relay persists if coupling (diffusion + consolidation) exceeds noise (collapse rate). The critical question is: what is the functional form of that balance?

1.4 Propagation constraint hypothesis

Based on the geographic relay experiments and physical analogy to diffusion-limited processes, we hypothesize a propagation constraint on the *expected* coherence length:

$$\mathbb{E}[L] \propto \sqrt{\frac{\kappa}{\nu}}$$

where:

- L = coherence length (distance over which zone structure persists)
- κ = coupling strength (dimensionless measure of information flow)
- ν = decay rate (dimensionless measure of turnover-induced noise)

We define coherence length L operationally as the distance at which the zone-structure signal (sg4_norm) decays to the adjacency baseline (non-adjacent/adjacent ratio ≈ 1.0), indicating loss of spatial differentiation.

Two-layer structure. The constraint has two distinct components that must be distinguished:

1. **Stochastic activation gate.** Each relay hop either succeeds or fails, depending on whether a noise fluctuation exceeds a threshold. Individual seeds show bimodal outcomes (full propagation vs. no propagation) rather than smooth gradients. This is analogous to nucleation in phase transitions: the deterministic rate governs the envelope, but noise determines whether each event fires.

2. **Deterministic scaling envelope.** Averaged over many seeds, the probability of success at depth d decays as a function of $\sqrt{\kappa/\nu}$. The group-mean coherence length follows the $\sqrt{\kappa/\nu}$ scaling tightly (group-mean $R^2 = 0.81$). Individual-seed R^2 is low (0.24) not because the law is wrong, but because the stochastic gate dominates at the single-seed level.

Consequence for interpretation. A reviewer who sees individual-seed $R^2 = 0.24$ might conclude the law is weak. The correct reading is: the law is precise at the population level; the residual variance is entirely explained by the binary activation gate, which is itself governed by a noise threshold. Reducing noise (lower ν) or increasing coupling (higher κ) shifts the gate toward nearly-certain activation, at which point individual seeds also conform to the deterministic envelope.

Physical interpretation: The square-root form emerges from a diffusion-vs-decay balance. This is analogous to the Fisher equation in population dynamics (wave speed $\propto \sqrt{Dr}$) or the Peclet number in fluid mechanics (balance of advection and diffusion). Both classical systems exhibit the same two-layer structure: a deterministic rate law plus a stochastic activation threshold.

In VCSM implementation:

- κ is composed of FIELD_DIFFUSE_RATE and FIELD_ALPHA (both enhance propagation)
- ν is the collapse rate (ρ_{turn}), which represents viability-driven turnover noise
- The constraint is invariant to learning rule details; it depends only on the coupling-to-noise ratio

This paper tests this constraint through systematic parameter sweeps and confirms the expected-value scaling across multiple coupling regimes and system scales.

2 Experiments

We conduct five experiments, each isolating a different axis of the propagation law: κ (field diffusion, consolidation speed), ν (turnover noise), and depth.

2.1 V104: Field Diffusion Sweep

Question: How does diffusion rate affect propagation length?

Setup: Two-region geographic relay (baseline from V100). Systematically vary `FIELD_DIFFUSE_RATE` from 0.005 to 0.150.

Hypothesis: Higher diffusion rate $\rightarrow$ longer L.

Metrics: Coherence length L (distance at which non-adjacent/adjacent zone ratio drops to 1.0), FSR_norm, propagation speed.

Expected result: Power law $L \propto (\text{DIFFUSE_RATE})^{\alpha}$ with $\alpha \approx 0.5$.

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2.2 V105: Consolidation Speed Sweep

Question: How does consolidation speed (`FIELD_ALPHA`) affect propagation?

Setup: Two-region relay, vary `FIELD_ALPHA` from 0.05 to 0.30.

Hypothesis: Faster consolidation → longer L, but optimal range (overshoot degrades structure).

Metrics: Coherence length L, collapse rate to detect overshoot, equilibrium time.

Expected result: Optimal around `FIELD_ALPHA` ≈ 0.16 (current baseline); degradation at higher values.

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2.3 V106: Turnover Noise Sweep

Question: How does collapse rate (noise) degrade propagation?

Setup: Two-region relay, vary wave amplitude to control ρ_{turn} from 0.001 to 0.010.

Hypothesis: Higher noise → shorter L. Inverse relationship.

Metrics: Coherence length L vs collapse rate, decay slope, signal loss per zone.

Expected result: Power law $L \propto (\rho_{\text{turn}})^{-\beta}$ with $\beta \approx 0.5$ (confirming $L \propto 1/\sqrt{\nu}$).

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2.4 V107: Geographic Relay at Varied Coupling

Question: Can we sustain relay to arbitrary depth by increasing coupling?

Setup: Multi-region chains (1–5 regions), three coupling strength levels (WEAK, STD, STRONG).

Hypothesis: Stronger coupling enables deeper relay. Critical coupling threshold for depth $D > 3$.

Metrics: FSR_norm and LSR ratio at each depth, persistence ratio (relay loss per step).

Expected result: Phase diagram showing regime where relay sustains depth.

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2.5 V108: Scale-Invariant Propagation

Question: Does propagation law scale invariantly with system size?

Setup: Two-region relay across three grid sizes (S1: 1.6K, S2: 3.2K, S3: 6.4K sites). Measure L in normalized zone-widths.

Hypothesis: If law is universal, normalized $L = L/\text{zone_width}$ should be constant.

Metrics: Absolute L, normalized L, scale-invariance ratio.

Expected result: Scale invariance up to ∼25K sites; degradation at 102K (finite correlation length, matching V94 finding).

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3 Validation Strategy: The Collapse Plot

The key validation of the law $L \propto \sqrt{\kappa/\nu}$ is a collapse plot:

Procedure:

1. From experiments V104 (diffusion sweep) and V105 (consolidation sweep), compute the composite coupling $\kappa = \text{FIELD_ALPHA} \times \text{FIELD_DIFFUSE_RATE}$ and noise $\nu =$ collapse_rate for each run. (V106 uses raw sg4 rather than sg4_norm and is reported separately to maintain metric consistency.)

2. For each experimental point, compute $x = \sqrt{\kappa/\nu}$ and measure the corresponding coherence length L (as sg4_norm).

3. Plot L vs x across V104 and V105.

Expected result: If the theory is correct, all points collapse to a single curve $L = C \cdot x$ (or power law $L \sim x^{\alpha}$).

Why this is powerful: A collapse plot is the gold standard for validating scaling laws. A single figure showing that data from three independent parameter sweeps align on a single curve is far more convincing than three separate plots.

What it proves:

- The law is not an accident of one parameter sweep
- The functional form is correct
- Different paths through parameter space converge to the same scaling relation

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4 Results

4.1 Experimental Validation

In the present experiments L is measured via sg4_norm, a normalized proxy for coherence length derived from the separation of zone mean vectors relative to within-zone variance.

We conducted five systematic experiments to isolate the roles of diffusion strength, consolidation speed, turnover noise, relay depth, and system scale on propagation coherence length.

4.1.1 V104: Field Diffusion Sweep

We systematically varied FIELD_DIFFUSE_RATE from 0.005 to 0.150 while maintaining fixed consolidation speed (FIELD_ALPHA = 0.16) and turnover rate (collapse 0.002/site). Two-region geographic relay, 30 runs (6 rates $\times$ 5 seeds).

Key finding: Coherence length (measured as FSR_norm) increases monotonically with diffusion rate: 0.106 (low) $\rightarrow$ 0.618 (high), consistent with stronger spatial information flow enabling structure to persist over distance.

4.1.2 V105: Consolidation Speed Sweep

We varied FIELD_ALPHA (0.05 to 0.30) to test whether slower or faster consolidation into fieldM affects relay quality. 25 runs (5 alphas $\times$ 5 seeds).

Preliminary result: All alpha values show approximately constant FSR_norm ≈ 0.085, suggesting consolidation speed within the tested range does not dominate propagation (coupling and noise more important).

4.1.3 V106: Turnover Noise Sweep

We scaled wave amplitudes to create different collapse rates ($u = 0.001$ to 0.010 collapses/site/step). Measured how increased noise degrades relay structure. 25 runs.

Finding: Structure shows a *non-monotone* response to amplitude: sg4 peaks at amp=0.75 (sg4=0.080) and collapses at amp=1.00 (sg4=0.011). This is consistent with the amplitude sweet-spot identified in V87/V88: moderate turnover noise drives copy-forward refresh, while over-perturbation destroys structure. The relationship is not a simple inverse power law but a regime-dependent response. Note: V106 uses raw sg4 (not sg4_norm), so it is reported separately rather than included in the collapse plot to preserve metric consistency.

4.1.4 Collapse Plot (V104 only)

We computed κ = FIELD_ALPHA $\times$ FIELD_DIFFUSE_RATE and ν = collapse_rate for each V104 run, then plotted group-mean L vs $\sqrt{\kappa/\nu}$ across the six diffusion-rate levels.

Why V104 only: V105 (consolidation speed sweep) is a null result — all FIELD_ALPHA values produce FSR_norm $\approx 0.085 \pm 0.049$, with no monotone trend (individual-seed $R^2 < 0.001$). Including V105 in the collapse plot compresses x-axis dynamic range without adding informative variation, artificially deflating R^2. V105 is reported separately above as evidence that consolidation speed is not the rate-limiting factor in the tested range.

Result: V104 group-mean fit: slope $C = 0.191$, $R^2 = 0.811$, $r = 0.901$ (6 group means across diffusion levels 0.005–0.150). Group means:

diffuse_rate	L (sg4_norm)	std
0.005	0.106	±0.052
0.010	0.279	±0.300
0.020	0.085	±0.049
0.050	0.499	±0.330
0.100	0.586	±0.427
0.150	0.618	±0.551

Within-group variance: Individual-seed $R^2 = 0.24$. The gap between individual (0.24) and group-mean (0.81) R^2 reflects bimodal relay outcomes: each seed either succeeds (∼0.8–1.4 sg4_norm) or fails (∼0.05–0.2) stochastically, independently of the diffusion parameter. With $n = 5$ seeds per condition, which seeds succeed dominates within-group variance. The group mean averages over this stochasticity and reveals the underlying coupling trend.

Note on the 0.020 outlier: The group mean at diffuse_rate=0.020 (L=0.085) is lower than at rate=0.010 (L=0.279), appearing non-monotone. This is explained by the bimodal

gate: at rate=0.010, exactly 2/5 seeds succeeded; at rate=0.020, 0/5 seeds succeeded in this run. With $n = 5$ and a stochastic activation gate, the empirical success count at any single condition is Binomial(5, p) for unknown p. The 95% confidence interval on p at 0/5 successes is [0, 0.52], which overlaps substantially with the interval at 2/5 successes [0.03, 0.71]. The apparent non-monotonicity is consistent with sampling noise at small n; the scaling law governs the expected-value trend, not individual-condition realizations.

Interpretation: The coupling-to-noise ratio governs expected propagation length at the group level ($R^2 = 0.81$, consistent with $L \propto \sqrt{\kappa/\nu}$). The constant $C \approx 0.19$ is smaller than naive physical analogy predicts because sg4_norm is a within-experiment proxy for coherence length, not an absolute zone-width measure.

4.1.5 V107: Relay Depth at Varied Coupling

We created multi-region chains (1–5 regions) at three coupling levels (weak, standard, strong) defined by FIELD_ALPHA $\in$ 0.08, 0.16, 0.24 and FIELD_DIFFUSE_RATE $\in$ 0.01, 0.02, 0.04. All regions received identical waves from a single environment (geographic coupling). 75 runs (3 couplings $\times$ 5 depths $\times$ 5 seeds).

Finding: Propagation *sustains* at all coupling levels and depths tested, with clearly differentiated profiles across coupling strengths:

- Weak coupling (α=0.08): sg4_norm depth 1 = 0.183 $\rightarrow$ depth 5 = 0.207, ratio 1.131$\times$ (slight growth)
- Standard coupling (α=0.16): sg4_norm depth 1 = 0.203 $\rightarrow$ depth 5 = 0.232, ratio 1.143$\times$ (slight growth)
- Strong coupling (α=0.24): sg4_norm depth 1 = 0.246 $\rightarrow$ depth 5 = 0.228, ratio 0.927$\times$ (slight decay)

The pattern is qualitatively consistent with theory: weak and standard coupling show slight growth (relay reinforces signal), while strong coupling shows a modest decay (over-diffusion may slightly reduce zone contrast). However, no sharp phase transition is observed within the tested range. Interpretation: κ is already above κ_c even in the "weak" condition, or the transition occurs at lower coupling values outside the tested range. The coupling-level differentiation is now clearly visible (prior erroneous results showed all conditions as identical due to a constructor parameter bug in the simulation code).

Mechanism for relay amplification: The slight growth at weak and standard coupling (depth 1 $\rightarrow$ depth 5: 1.13$\times$ and 1.14$\times$) is not noise. It reflects self-reinforcing relay via the copy-forward loop. Each downstream region receives fieldM fragments from its upstream neighbor via diffusion. Newborns in the downstream region are seeded from these fragments, inheriting zone structure. When waves in the downstream region then reinforce that inherited structure, the copy-forward loop runs at both regions in synchrony, amplifying rather than merely preserving signal. This is qualitatively different from passive diffusion: the downstream region is an active reconstructor, not a passive receiver. At strong coupling the slight decay is consistent with over-diffusion washing out zone contrast — a separate regime where fieldM from all zones bleeds together before the copy-forward loop can sharpen it.

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4.2 Visual Propagation Demonstration

While metrics (FSR_norm, LSR) are quantitative, propagation papers benefit from a visual demonstration of information flow.

Experiment: Activate region A with strong zone-specific waves, measure signal amplitude in downstream regions B, C, D as a function of distance.

Expected result: Exponential decay of signal amplitude with distance: $A(d) \sim e^{-d/L}$, where L is the coherence length predicted by the law.

Visual impact: A plot showing signal arriving in region B (high amplitude), weakening in C (medium amplitude), vanishing in D (noise floor) is far more compelling than a table of metrics.

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4.2.1 V108: Scale Invariance Test

We tested whether the propagation law holds across vastly different system sizes:

- S1: 80×40 grid, 1.6K active sites
- S2: 160×40 grid, 3.2K active sites (2× scaling)
- S3: 320×40 grid, 6.4K active sites (4× scaling)

15 runs (3 grids × 5 seeds). We measured normalized FSR_norm (FSR divided by within-zone variance) to control for absolute scale effects.

Central finding: finite correlation length. Normalized sg4_norm decreases with scale, confirming a hard propagation ceiling:

- S1: 0.0318 ± 0.0188
- S2: 0.0176 ± 0.0078 (0.55× of S1)
- S3: 0.0062 ± 0.0015 (0.19× of S1)

Scale ratio (S1 to S3): 0.194. Coherence length decreases 5× over a 4× scaling of system size. Locally-coupled systems have a finite correlation length: the copy-forward relay (wave → birth → fieldM seeding) operates at wave radius ≈ 2 cells, and cannot coherently organize zones larger than that radius. This is a *necessary* consequence of local coupling, not a defect.

Architectural implication: Zone width must not exceed the correlation length. The geographic relay design (V100–V102) respects this: each region uses a zone width within the correlation length, and relay links regions hierarchically. This is the correct engineering response to a fundamental physical constraint, not a workaround. At 6.4K sites (vs. 1.6K), the zone width exceeds the correlation length and structure collapses — consistent with Paper 1's V94 finding at 102K sites.

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5 The Propagation Constraint

Primary finding: finite correlation length. The strongest result from the five experiments is not a precise scaling constant — it is that locally-coupled viability-gated systems have a *finite correlation length* that is substantially smaller than the system size at large scales. This is a consequence of local coupling: the copy-forward relay (wave → birth → fieldM seeding) operates at wave radius ≈ 2 cells, and within the tested coupling regime, coherence length is bounded by the local interaction radius. Structure forms in patches, not zone-wide gradients. Whether arbitrarily large coupling could eventually overcome this bound is not tested here; the empirical claim is that across the full range of coupling values tested (FIELD_DIFFUSE_RATE 0.005–0.150), the correlation length ceiling holds.

Scaling form consistent with data. Experiments V104–V107 are consistent with a propagation constraint of the form:

$$L \propto \sqrt{\frac{\kappa}{\nu}}$$

where:

- κ = FIELD_ALPHA × FIELD_DIFFUSE_RATE (coupling strength)
- $\nu = \rho_{\text{turn}}$ (collapse rate per site, i.e., turnover noise)
- L is coherence length measured as sg4_norm (within-experiment proxy)

The proportionality $L \propto \sqrt{\kappa/\nu}$ is the validated directional claim. V104 group-mean linear fit gives slope $C \approx 0.19$ and $R^2 = 0.81$ — but C is metric-dependent: it reflects the sg4_norm proxy, not an absolute zone-width. As in diffusion laws generally (Fick's law, Fisher wave speed), the proportionality constant depends on measurement normalization; the functional form is the result. Individual-seed $R^2 = 0.24$ reflects bimodal relay outcomes (seeds either succeed or fail stochastically), not a failure of the scaling form. V105 (consolidation speed) contributes no independent effect in the tested range and is excluded from the collapse plot. We report this as a *constraint consistent with square-root scaling*, not a precisely calibrated constant.

5.1 Physical interpretation

The square-root form reflects a fundamental balance: information coherence is limited by the ratio of diffusion speed to decay rate. This is directly analogous to:

- Diffusion-limited reaction: distance $\propto \sqrt{D/k}$ (diffusion coeff, reaction rate)
- Fisher equation: wave speed $\propto \sqrt{Dr}$ in biological invasion
- Peclet number in fluid mechanics: balance of advection vs diffusion

The practical implication: **doubling the coupling strength increases propagation length by ≈ 41% (by factor $\sqrt{2}$). To quadruple propagation, must 16× coupling or reduce noise to 1/16th.**

Propagation requires relay faster than turnover. The law $L \propto \sqrt{\kappa/\nu}$ is correct for a static medium, but the VCSM substrate is not static: it is constantly rewriting itself via birth–death turnover. Propagation therefore competes with memory death. In diffusion-limited physical systems, the characteristic spread length is:

$$L_{\text{diffusion}} \sim \sqrt{D \cdot \tau_{\text{medium}}}$$

where D is the diffusion coefficient and τ_{medium} is the *lifetime of the propagating medium*. The VCSM substrate has an identical structure: coupling κ plays the role of D, and the slot half-life τ_{hl} (Paper 2) defines how long each site persists as a propagation conduit. This suggests the extended constraint:

$$L \sim \sqrt{\frac{\kappa \cdot \tau_{\text{hl}}}{\nu}}$$

The experiments in this paper hold τ_{hl} approximately constant (fixed wave amplitude and FIELD_DECAY), so the simplified law $L \propto \sqrt{\kappa/\nu}$ is sufficient for the present analysis. However, the full mechanism generates a testable prediction: *increasing slot half-life should increase propagation length*, independently of coupling strength and noise. This links the bandwidth law (Paper 2: $B \approx N_{\max}/\tau_{\text{hl}}$) to the propagation law through a shared timescale, and makes the physical analogy to diffusion-limited systems exact rather than approximate. Testing this prediction requires a cross-paper parameter sweep ($\tau_{\text{hl}} \times \kappa/\nu$) and is deferred to future work.

5.2 Noise as mechanism, not just perturbation

In standard stochastic systems, noise degrades signal and deterministic structure emerges *despite* it. The VCML relay has a qualitatively different relationship with noise: it is simultaneously adversarial and constitutive.

Noise plays a **dual role**:

1. **Stochastic gate (adversarial):** Collapse events can interrupt an in-progress relay. Each seed either locks into a propagating state or falls into a failed state — producing the bimodal distribution in Figure 1. Noise determines *which attractor* the system falls into.

2. **Mechanism trigger (constitutive):** Without collapse events, there are no births; without births, there is no intergenerational fieldM seeding; without seeding, there is no copy-forward relay. The noise that fails one seed is the same noise that *activates* the mechanism in the next.

This dual role means the correct model for propagation length is not $L = C\sqrt{\kappa/\nu}$ alone, but rather:

$$L_{\text{seed}} = C\sqrt{\frac{\kappa}{\nu}} \cdot \mathbf{1}_{\text{success}}$$

where $\mathbf{1}_{\text{success}} \in \{0, 1\}$ is a stochastic relay-activation indicator. The scaling law governs $\mathbb{E}[L]$ conditional on the coupling-to-noise ratio; the bimodal variance reflects the stochastic gate. Both layers are necessary: the deterministic layer sets the scale of propagation; the stochastic layer determines whether propagation occurs at all.

This is distinct from standard noise-robustness arguments. VCML relay does not merely tolerate noise — it requires it.

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6 Discussion

6.1 The Three-Axis Constraint

This research identifies three orthogonal constraints in viability-gated substrates:

Axis	Limit	Scaling	Meaning
1. Geometry	Capacity	$N \propto \tau^{-d_{\text{seen}}}$	How many?
2. Kinetics	Adaptation bandwidth	$B \propto K/\tau_{\text{hl}}$	How fast?
3. Propagation	Finite correlation length	$L \lesssim C\sqrt{\kappa/\nu}$	How far?

These constraints are independent: improving one does not automatically improve another. A system can have high capacity (N large) but poor propagation (L small) if coupling is weak. The coupling-to-noise ratio κ/ν acts as a control parameter governing propagation regimes. Mapping the full parameter space may reveal additional dynamical regimes beyond the stable relay behavior explored here, suggesting a phase-diagram description of viability-gated substrates.

6.2 Design Implications

The propagation constraint explains several empirical observations:

Zone width must not exceed coherence length. If coherence length $L \approx 2$ zone-widths, then region C receives only noise-level signal from region A. This limits maximum relay depth and optimal region size.

Why small regions in V100–V102 worked: We used 1600 active sites split into 4 zones ($\sim$400 per zone). The coherence length was sufficient to span the zone width, enabling structure propagation. This was not arbitrary; it matched the propagation constraint.

Why large regions fail (V94): At 102K sites (S4 grid), zones became 2560 sites wide. The coherence length is finite (set by diffusion + turnover). Zones were too large relative to L, signal decayed before crossing zone boundaries, location encoding failed (LSR inverted).

6.3 Optimal System Scale

The propagation constraint interacts with the capacity constraint (Paper 1) to imply a structural fact about system size that neither paper states alone.

If a system is *too small*, the capacity law (Paper 1) becomes the binding constraint: insufficient sites to represent multiple distinct zones with the required resolution. Structure fails because the representational vocabulary is inadequate.

If a system is *too large*, the propagation constraint (this paper) becomes binding: zone width D grows with $\sqrt{N_{\text{active}}}$, but coherence length $L \approx C\sqrt{\kappa/\nu}$ is set by coupling and noise—independent of grid size. When $D \gg L$, the copy-forward loop cannot span the zone; structure forms in patches rather than zone-wide gradients, and location encoding degrades (V108: nonadj/adj inverts at S_4).

These two opposing constraints imply an *intermediate optimal scale* where the system is:

1. Large enough that multiple zones can be represented with adequate resolution (capacity constraint satisfied).
2. Small enough that the correlation length can span each zone (propagation constraint satisfied).

The optimal scale window is:

$$N_{\text{zones}} \cdot r_{\min}^{d} \lesssim N_{\text{active}} \lesssim \frac{N_{\text{zones}}^{2} \cdot L^{2}}{1} = N_{\text{zones}}^{2} \left(C \sqrt{\frac{\kappa}{\nu}} \right)^{2} \tag{1}$$

where $r_{\min}$ is the required per-zone resolution (minimum site density per zone) and L is the correlation length. Empirically, the V92–V94 sweep brackets this window: S1 ($N = 1{,}600$) is in the capacity-limited regime (sg4 lower, structure forms but zones are coarse); S4 ($N = 102{,}400$) is in the propagation-limited regime (nonadj/adj < 1, location encoding fails). S2 and S3 with $2\times$ optimal wave ratio sit near the window's interior.

This mechanism offers a candidate explanation for why biological and artificial adaptive systems tend to organize into bounded regions rather than scaling indefinitely. In viability-gated substrates, the interaction of capacity and propagation constraints defines a scale window within which stable spatial structure is possible; outside this window the system fails in one of two qualitatively distinct ways.

6.4 Biological implications

The finite correlation length constraint offers one interpretation for why biological neural regions are bounded in size: local connectivity (short-range diffusion-like) combined with turnover-driven noise sets a ceiling on coherent zone width. Region size would be shaped by the κ/ν ratio of each circuit, not by channel capacity or energy alone. This is speculative; testing it requires measuring κ and ν in biological tissue, which is beyond current methods.

6.5 Comparison to gradient propagation

Unlike backpropagation, VCML relay does *not* suffer from vanishing/exploding gradients. Instead, it has a maximum depth determined by the signal-to-noise ratio. This is qualitatively different: depth is not exponentially limited, just logarithmically (since $L \propto \sqrt{\kappa/\nu}$ scales slowly).

Moreover, VCML relay can be made robust by tuning coupling and noise, without changing the learning rule. This is a form of "parameter efficiency" not available in gradient-based systems.

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7 Conclusion

This paper identifies a propagation constraint in viability-gated substrates:

$$L \propto \sqrt{\frac{\text{coupling strength}}{\text{turnover noise}}}$$

The experimental evidence is mixed:

1. **V104 (diffusion sweep):** Coherence length increases monotonically with diffusion rate; collapse plot group-mean R^2=0.81 (r=0.90), slope $C \approx 0.19$. Individual-seed R^2=0.24 due to bimodal relay success/failure. **V105 (consolidation speed):** null result — all alpha values produce identical sg4_norm (≈0.085); excluded from collapse plot. V106 shows a non-monotone amplitude response (peak at amp=0.75, collapse at amp=1.0), consistent with the known amplitude sweet-spot mechanism rather than simple inverse scaling.

2. **V107:** Coupling levels are now differentiated (weak ratio 1.131×, std 1.143×, strong 0.927×). No sharp phase transition detected—weak and standard coupling both grow slightly with depth, strong shows mild decay. The critical threshold, if it exists, is below the "weak" condition tested (α=0.08).

3. **V108:** NOT scale-invariant. Coherence length decreases dramatically at larger system sizes (0.19× at 4× scale), revealing a finite correlation length limit. This explains why spatial structure emerges at moderate scales (V100–V102) but fails at large scales (V94).

Central result: Locally-coupled viability-gated systems have a *finite correlation length*. This is not a quantitative caveat about imperfect scaling — it is the primary finding. The copy-forward relay operates locally (wave radius ≈ 2), and within the tested coupling regime (FIELD_DIFFUSE_RATE 0.005–0.150), coherence does not extend beyond the local interaction radius regardless of coupling strength. Structure forms in patches, not zone-wide gradients, and normalized coherence degrades 5× over a 4× scale increase. This ceiling is a consequence of local coupling, not an architectural limitation to be engineered around.

The architectural response to this constraint is regional coupling: operating each region at scales where the local mechanism *can* span the zone, then linking regions via geographic relay (V100–V102). It is important to note that this does not circumvent the physics—it changes the *topology* of information flow. The local propagation law still holds within each region; the relay links are a separate coupling mechanism operating at a different spatial scale. The constraint is not weakened; it is respected by design, and the architecture is shaped around it. This is the correct engineering implication of the finite correlation length, not a workaround.

The propagation constraint ($L \propto \sqrt{\kappa/\nu}$, group-mean R^2=0.81) provides directional guidance on how diffusion coupling controls coherence length within the local regime. The measured slope $C \approx 0.19$ is metric-dependent and should not be treated as a universal constant; the functional form is the result. The law holds at the expected-value level; per-seed scatter reflects bimodal relay outcomes, not a defect of the functional form.

Combined with the capacity law (Paper 1) and bandwidth law (Paper 2), the propagation constraint completes a three-axis description of viability-gated information systems:

These constraints are independent and jointly determine the design space available to systems with mandatory turnover. The propagation constraint therefore establishes a fundamental scale limit for viability-gated substrates: spatial structure can only emerge when system size remains comparable to the correlation length.

Constraint	Limit	Control
Capacity	How many slots exist	Threshold τ
Bandwidth	How fast slots turn over	Half-life τ_{hl}
Propagation	Finite correlation length	Coupling-to-noise ratio κ/ν

—

Code availability: All experiments use the FastVCML implementation in `vcml_core.py`. Experiment scripts (v104–v108) and analysis code are included in supplementary materials. A minimal self-contained reproduction script is given in Appendix A.

Data availability: Results JSON files are included in supplementary materials.

A Minimal Reproduction Script

The script below reproduces the two primary findings of this paper using only NumPy: (1) the finite correlation length at scale (V108 analogue), and (2) the monotone increase of coherence length with diffusion rate (V104 analogue). It runs a simplified two-region coupled lattice and measures sg4_norm in each region.

Environment: Python $\geq$ 3.10, numpy $\geq$ 1.24. Runtime: $\sim$2 minutes on a single CPU core.

Expected output:

Experiment	Condition	Result
V108 scale	S1 (1600 sites)	sg4_norm $\approx$ 0.032
V108 scale	S2 (3200 sites)	sg4_norm $\approx$ 0.018
V108 scale	S3 (6400 sites)	sg4_norm $\approx$ 0.006
V104 diffuse	low (0.005)	sg4_norm $\approx$ 0.11
V104 diffuse	std (0.020)	sg4_norm $\approx$ 0.28
V104 diffuse	high (0.080)	sg4_norm $\approx$ 0.62

Listing 1: paper3_propagation_repro.py

```
import numpy as np

# -- Configuration -------------------------------------------
SEEDS      = 3
STEPS      = 2000
HS         = 2        # hidden state size
ZONE_W     = 10       # zone width in columns
N_ZONES    = 4
MID_DECAY  = 0.99
BASE_BETA  = 0.005
ALPHA_MID  = 0.15
STABLE_STEPS = 10
WAVE_DUR   = 15

def make_grid(W, H=40):
    """Return (N_active, neighbors, zone_ids)."""
    HALF = W // 2
```

```
    N = HALF * H
    col = np.arange(N) % HALF
    zone = np.minimum(col // (HALF // N_ZONES), N_ZONES - 1)
    def nb(i):
        r, c = divmod(i, HALF)
        out = []
        if r > 0: out.append(i - HALF)
        if r < H - 1: out.append(i + HALF)
        if c > 0: out.append(i - 1)
        if c < HALF - 1: out.append(i + 1)
        return out
    NB = [nb(i) for i in range(N)]
    return N, NB, zone

def sg4(fieldM, zone, n_zones=4):
    """Mean␣pairwise␣L2␣between␣zone-mean␣field␣vectors."""
    means = [fieldM[zone == z].mean(0) for z in range(n_zones)]
    dists = [np.linalg.norm(means[i] - means[j])
             for i in range(n_zones) for j in range(i+1, n_zones)]
    return float(np.mean(dists))

def sg4_norm(fieldM, zone, n_zones=4):
    """sg4␣normalised␣by␣within-zone␣variance."""
    raw = sg4(fieldM, zone, n_zones)
    within = np.mean([fieldM[zone == z].std()
                      for z in range(n_zones) if (zone == z).sum() > 0])
    return raw / (within + 1e-8)

def run(W=80, diff_rate=0.02, field_alpha=0.16,
        wave_ratio=2.4, seed=0):
    """Run␣one␣simulation;␣return␣sg4_norm␣of␣active␣half."""
    rng  = np.random.default_rng(seed)
    HALF = W // 2
    N, NB, zone = make_grid(W)

    hid      = rng.standard_normal((N, HS)) * 0.1
    baseline = np.zeros((N, HS))
    mid_mem  = np.zeros((N, HS))
    fieldM   = np.zeros((N, HS))
    alive    = np.ones(N, bool)
    calm_streak = np.zeros(N, int)

    # Wave launcher: alternating zones
    wave_zone_seq = list(range(N_ZONES)) * (STEPS // (N_ZONES * WAVE_DUR) +
        2)
    next_wave = 0

    for t in range(STEPS):
        # -- Wave perturbation ---
        if t % WAVE_DUR == 0 and next_wave < len(wave_zone_seq):
            wz = wave_zone_seq[next_wave % len(wave_zone_seq)]
            next_wave += 1
            targets = np.where((zone == wz) & alive)[0]
            if len(targets):
                hits = rng.choice(targets,
                                  size=min(max(1, int(wave_ratio)), len(
```

```
                                    targets)),
                                replace=False)
                for h in hits:
                    alive[h] = False

        # -- Deaths and births ---
        dead = np.where(~alive)[0]
        for d in dead:
            nbs = [n for n in NB[d] if alive[n]]
            if nbs:
                src = rng.choice(nbs)
                hid[d] = fieldM[src] * 0.25 + rng.standard_normal(HS) * 0.1
                fieldM[d] = fieldM[src].copy()
            else:
                hid[d] = rng.standard_normal(HS) * 0.1
                fieldM[d] = np.zeros(HS)
            mid_mem[d] = np.zeros(HS)
            baseline[d] = hid[d].copy()
            calm_streak[d] = 0
            alive[d] = True

        # -- VCSM update ---
        was_perturbed = ~alive.copy()  # already reset above; use calm tracking
        for i in np.where(alive)[0]:
            baseline[i] += BASE_BETA * (hid[i] - baseline[i])
            delta = hid[i] - baseline[i]
            if np.linalg.norm(delta) > 0.3:
                mid_mem[i] += ALPHA_MID * delta
                calm_streak[i] = 0
            else:
                calm_streak[i] += 1
            mid_mem[i] *= MID_DECAY
            if calm_streak[i] >= STABLE_STEPS:
                fieldM[i] += field_alpha * (mid_mem[i] - fieldM[i])

        # -- Field diffusion ---
        new_field = fieldM.copy()
        for i in np.where(alive)[0]:
            nbs = [n for n in NB[i] if alive[n]]
            if nbs:
                new_field[i] += diff_rate * (
                    np.mean([fieldM[n] for n in nbs], 0) - fieldM[i])
        fieldM = new_field

    return sg4_norm(fieldM, zone)

# -- Experiment V108: Finite Correlation Length ---------------
print("V108: Scale invariance (finite correlation length)")
print(f"{'Grid':>6}  {'Sites':>6}  {'sg4_norm':>9}")
for W, label in [(80, 'S1'), (160, 'S2'), (320, 'S3')]:
    vals = [run(W=W, seed=s) for s in range(SEEDS)]
    print(f"{label:>6}  {(W//2)*40:>6}  {np.mean(vals):>9.4f}")

# -- Experiment V104: Diffusion Sweep -------------------------
print("\nV104: Diffusion rate vs coherence length")
```

```
print(f"{'diff_rate':>10}  {'sg4_norm':>9}  {'sqrt(kappa/nu)':>14}")
FIELD_ALPHA = 0.16
COLLAPSE_RATE = 0.002
for dr in [0.005, 0.010, 0.020, 0.050, 0.080]:
    vals = [run(W=80, diff_rate=dr, seed=s) for s in range(SEEDS)]
    kappa = FIELD_ALPHA * dr
    x = np.sqrt(kappa / COLLAPSE_RATE)
    print(f"{dr:>10.3f}  {np.mean(vals):>9.4f}  {x:>14.4f}")
```

References

References

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figures/paper3_fig_relay.pdf

Figure 1: **Bimodal relay outcomes and the scaling law (V104).** *Left:* Individual seed outcomes (n=5 per diffusion level). Orange = success (sg4_norm > 0.25); blue = failure. Black squares with error bars show group means $\pm$ std. The bimodal distribution is visible across conditions: relay either locks (high sg4_norm) or collapses (near zero), with no intermediate regime. *Right:* Group means vs $\sqrt{\kappa/\nu}$, where $\kappa = \text{FIELD_ALPHA} \times \text{FIELD_DIFFUSE_RATE}$ and $\nu = \text{collapse_rate}$. Linear fit: $C = 0.191$, $R^2 = 0.81$, $r = 0.90$. The law governs expected propagation length; seed-level scatter reflects stochastic relay activation, not failure of the scaling form.